

NATHÃ SOUZA

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PROFESSIONAL SUMMARY

AI Engineer in the final stage of an M.S. in Artificial Intelligence, bridging a Chemical Engineering background with hands-on machine learning engineering. Experienced in building end-to-end, production-oriented ML systems: no-leakage walk-forward validation, probability calibration, live API data pipelines, and automated deployment. Strong foundation in process optimization and quantitative analysis from industrial R&D.

TECHNICAL SKILLS

- Machine Learning: XGBoost, scikit-learn, isotonic calibration, walk-forward validation, time-series modeling, PyTorch, transfer learning (MobileNetV2)
- Programming & Data: Python (pandas, NumPy), SQL, Git, REST API integration, JSON, automation scripting (PyAutoGUI)
- Platforms & Tools: Google Colab, Binance API, Coinbase Advanced Trade API, Telegram Bot API, Streamlit
- Process & Quality Engineering: Aspen Plus, MATLAB, ChemDraw, Statistical Process Control (SPC), Design of Experiments
- Analytical Instrumentation: Spectroscopy (UV-Vis, FTIR), Chromatography (HPLC, GC)

AI / ML PROJECTS

ML-Based Algorithmic Trading Signal System

Personal Project

- Designed and deployed a no-leakage, walk-forward machine learning pipeline generating BTC/USDT M15 trading signals, using XGBoost with isotonic probability calibration to replace a degraded rule-based system.
- Built automated data ingestion from the Binance API and a five-stage Python pipeline covering feature engineering, model training, calibration, and signal generation.
- Implemented live signal delivery via a Telegram bot and automated trade execution using image-based, coordinate-corrected desktop automation (PyAutoGUI).

Algorithmic Trading Bot - Coinbase Advanced Trade API

Personal Project

- Rebuilt a Python trading bot and backtesting framework, correcting statistical flaws including RSI/ATR calculation (Wilder RMA), Sharpe ratio annualization, and fee/slippage modeling.
- Added a macro trend filter (MA200) with a configurable CLI ablation flag, and implemented JSON-based state persistence for reliability.

Transfer Learning for Image Classification

Academic — M.S. in AI

- Fine-tuned a MobileNetV2 architecture on CIFAR-10 in PyTorch on Google Colab, evaluating performance across training configurations under GPU resource constraints.

WORK HISTORY

Chemical Engineering Intern

07/2022 – 07/2024

BASF SE — Guaratinguetá, SP

- Optimized a fungicide formulation by evaluating alternative surfactants, reducing production costs by ~10% while maintaining product stability.
- Applied Python for data analysis to support process optimization strategies, contributing to a 20% increase in operational efficiency.
- Implemented a standardized product quality testing procedure, improving lab efficiency and reducing errors.
- Conducted chemical analyses (UV-Vis, FTIR, HPLC, GC) to support production processes and ensure product quality.
- Collaborated with cross-functional teams to troubleshoot technical issues and maintain laboratory equipment and safety protocols.

EDUCATION

M.S. in Artificial Intelligence

Expected 2027

Atlantis University — Florida

B.S. in Chemical Engineering

12/2024

Universidade de São Paulo (USP) — Lorena, SP. Selected to present final project to a panel of professors and industry professionals.

Technical Degree, Chemistry

12/2017

ETECAP

LANGUAGES

Portuguese — Native | English — Full Professional | Spanish — Limited Working